Appendix

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A Derivations for the Theoretical Framework

This appendix provides derivations for the value comparisons discussed in Section (2). To keep the main text focused on intuition and empirical predictions, the algebraic comparisons between telework, office work, and no paid work are collected here.

A.1 Telework versus Office Work

Telework is preferred to office work if

$$V_T > V_O. \tag{1}$$

For expositional simplicity, suppose that both work options require the same hours L. Then the value of telework can be written as

$$V_T = u(y_h + w_T L, \ell_T, f(d_T)) - \tau_T v(L) - \phi z(d_T), \tag{2}$$

where

$$\ell = 1 - L - d_T. \tag{3}$$

The value of office work is

$$V_O = u(y_h + w_O L, \ell_O, f(d_O)) - \tau_O v(L), \tag{4}$$

where

$$\ell = 1 - (1 + t_O)L - d_O. \tag{5}$$

Using a first-order approximation around a common allocation gives

$$V_T - V_O \approx u_c(w_T - w_O)L + u_H\{f(d_T) - f(d_O)\} + (\tau_O - \tau_T)v(L) - \phi z(d_T). \tag{6}$$

If the domestic-time allocation is held fixed for the comparison, then $d_T = d_O$, and the household-production term drops out. Since $\ell_T - \ell_O = t_O L$, telework is preferred to office work if

$$u_c(w_T - w_O)L + u_\ell t_O L + (\tau_O - \tau_T)v(L) - \phi z(d_T) > 0. \tag{7}$$

Rearranging yields

$$w_O - w_T < \frac{u_\ell}{u_c} t_O + \frac{(\tau_O - \tau_T)v(L)}{u_c L} - \frac{\phi z(d_T)}{u_c L}. \tag{8}$$

A.2 Telework versus No Paid Work

Telework is preferred to no paid work if

$$V_T > V_N. \tag{9}$$

The value of no paid work is

$$V_N = u(y_h, \ell, f(d_N)). \tag{10}$$

Using a first-order approximation,

$$V_T - V_N \approx u_c w_T L + u_H\{f(d_T) - f(d_N)\} - \tau_T v(L) - \phi z(d_T). \tag{11}$$

Therefore, telework is preferred to no paid work if

$$u_c w_T L > \tau_T v(L) + \phi z(d_T) + u_H\{f(d_N) - f(d_T)\}. \tag{12}$$

A.3 Office Work versus No Paid Work

Office work is preferred to no paid work if

$$V_O > V_N. \tag{13}$$

Using a first-order approximation,

$$V_O - V_N \approx u_c w_O L + u_H\{f(d_O) - f(d_N)\} - \tau_O v(L). \tag{14}$$

Since office work requires both paid work time and commuting time, the leisure loss relative to no paid work is approximately

$$\ell_N - \ell_O = (1 + t_O)L. \tag{15}$$

Thus, office work is preferred to no paid work if

$$u_c w_O L > \tau_O v(L) + u_\ell(1 + t_O)L + u_H\{f(d_N) - f(d_O)\}. \tag{16}$$

Equivalently,

$$w_O > \frac{\tau_O v(L)}{u_c L} + \frac{u_\ell}{u_c}(1 + t_O) + \frac{u_H\{f(d_N) - f(d_O)\}}{u_c L}. \tag{17}$$

A.4 Comparative Statics

The model implies that an increase in telework exposure raises the relative value of telework by reducing the effective time and mobility cost of paid work. Let F denote telework flexibility. A higher value of F can reduce commuting or mobility costs, so that

$$\frac{\partial t_T}{\partial F} < 0, \qquad \frac{\partial \tau_T}{\partial F} < 0. \tag{18}$$

Since telework has $t_T = 0$ in the baseline setup, the relevant effect of greater telework exposure is to increase the feasibility and attractiveness of the telework option itself. Therefore,

$$\frac{\partial (V_T - V_N)}{\partial F} > 0 \tag{19}$$

whenever flexibility reduces the cost of working more than it increases home-based work frictions. Similarly,

$$\frac{\partial (V_T - V_O)}{\partial F} > 0 \tag{20}$$

when telework reduces commuting, mobility, or workplace-related costs relative to office work.

These comparative statics motivate the empirical analysis: districts with greater telework exposure should experience larger increases in women's paid employment, especially among women for whom office work is more costly because of mobility restrictions, household responsibilities, or commuting constraints.

B Figures

B.1 Comparison between LLM-generated and text classifier approach

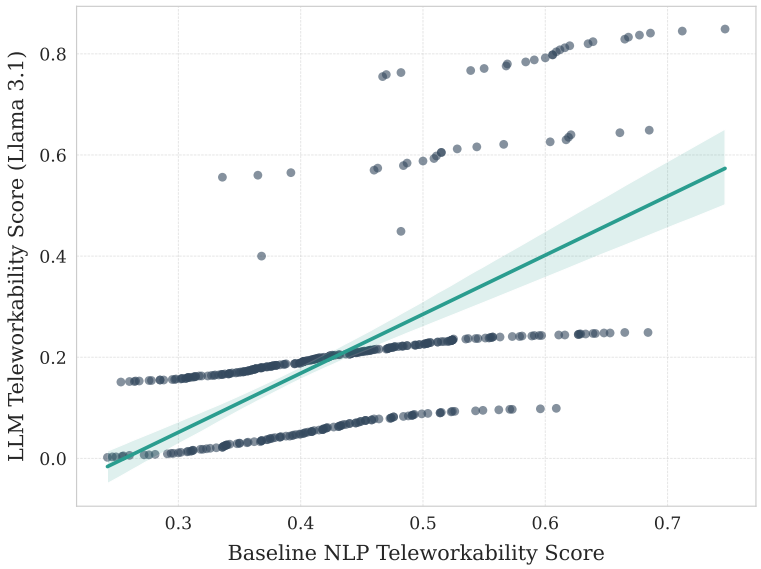
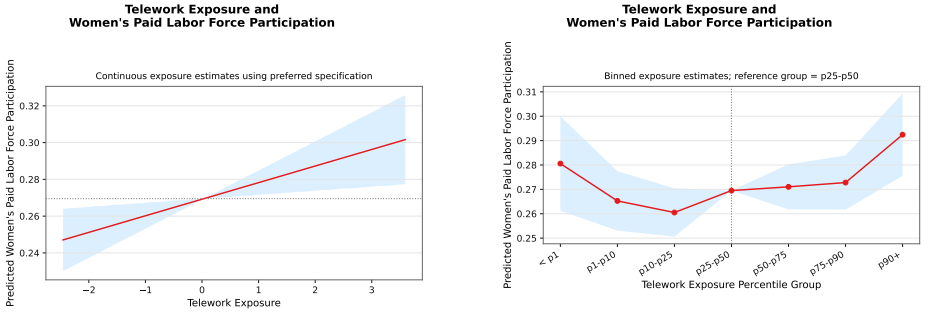


Figure 1: Comparison between NLP and LLM generated telework scores.

B.2 Plots of predicted labor supply by telework exposure



(a) Continuous exposure

(b) Binned exposure

Figure 2: Predicted Effects of Telework Exposure on Female Paid Labor Supply

Notes: Panels (a) and (b) plot predicted female paid labor participation across the continuous and binned distributions of telework exposure, respectively. In Panel (b), the 25th–50th percentile bin is the reference category. Shaded bands denote 95% confidence intervals. Standard errors are clustered at the district level.

Panel 2a shows a positive, monotonically increasing gradient: predicted female paid labor participation rises from approximately 20% at the lowest levels of telework exposure to around 39% at the highest levels. Panel 2b shows that the positive association persists when exposure is modeled flexibly using percentile bins, although the relationship is not monotonic at the lower end of the distribution. Predicted paid labor participation initially declines from below the 1st percentile to the 10th–15th percentile range, and subsequently increases as telework exposure rises. Higher-exposure districts include Northwest Delhi, Chennai, Bangalore, Pune, and Thane, whereas districts at or below the 25th percentile include Ahmedabad, Nadia, Jaipur, Murshidabad, Nagpur, and Chittoor.

B.3 Hypothetical Transmitter Lattice

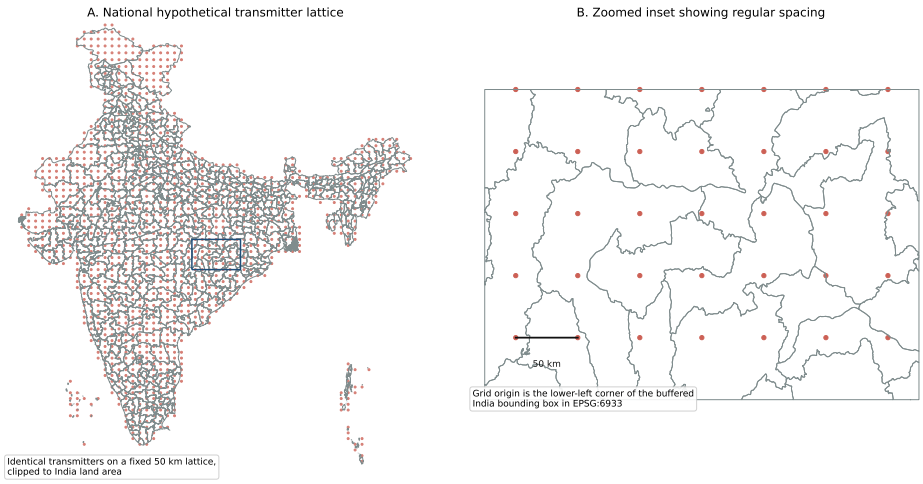


Figure 3: Distribution of hypothetical transmitters across India

C Tables

C.1 Results on the Locked Test Set

Table 1: Locked-Test Performance at Alternative Classification Thresholds

	Threshold	Bal. Acc.	F1	ROC–AUC	PR–AUC
F1-optimal	**0.468**	0.814	0.714	0.902	0.821
Youden-J optimal	**0.468**	0.814	0.714	0.902	0.821

Notes: The F1-optimal and Youden's J-optimal thresholds are selected using out-of-fold predictions from the 250-observation training sample. Both criteria select $\tau = 0.468$. The table reports classification performance on the independent locked test set of 150 occupations. At the training-selected threshold, balanced accuracy is 0.814 and the F1 score for the teleworkable class is 0.714. ROC–AUC and PR–AUC are threshold-independent and therefore remain unchanged across rows. The locked test set is not used for threshold selection.

Table 2: Performance of Text Classifier on the Locked Test Set

	Precision	Recall	F1	N
1 – Teleworkable	0.63636	0.81395	**0.71429**	43
0 – Not teleworkable	0.91579	0.81308	**0.86139**	107
Accuracy			0.81333	150
Macro Average	0.77608	0.81352	**0.78784**	150
Weighted Average	0.83569	0.81333	**0.81922**	150

Notes: This table reports classification performance on the independent locked test set of 150 manually labeled occupations. The classifier is trained on 250 separate manually labeled occupations, and the threshold $\tau = 0.46752$ is selected using out-of-fold predictions from the training sample only. The locked test set is used solely for final validation. See Appendix D.2 for details on the NLP terminology.

C.2 LLM-generated Telework Scores

Table 3: Top 10 and bottom 10 occupations by LLM-generated telework scores

Code	Occupation Title	LLM-generated score	NLP-generated score
Panel A: Top 10 by LLM-generated telework score			
2511.0102	Engineer-Software Transition	1.000	0.695
2412.0200	Financial Analyst	0.988	0.630
2641.0801	Associate-Learning	0.975	0.665
1330.0101	Communication Analyst	0.962	0.615
2519.0501	Quality Engineer	0.950	0.694
3118.0200	Draughtsperson, Civil	0.938	0.417
3521.0511	Sound Editor	0.925	0.597
2512.0205	Junior Software Developer	0.912	0.630
2641.0602	Script Researcher	0.900	0.565
2641.0302	Script Editor	0.888	0.547
Panel B: Bottom 10 by LLM-generated telework score			
8152.0700	Sock Knitter/Knitting Machine	0.002	0.232
7523.1100	Tenoning Machine Operator	0.003	0.226
8131.7000	Tablet Machine Operator	0.005	0.228
8157.0100	Washing Machine Operator	0.006	0.218
8160.8700	Cutting Machine Operator, Tobacco	0.008	0.238
8171.0100	Chipper Man, Paper Pulp	0.010	0.246
8112.0700	Stone Sawyer, Machine	0.011	0.225
8131.4200	Box Making Machine Operator	0.013	0.242
7233.0800	Mechanic, Road Roller	0.015	0.252
8342.0400	Grader Operator	0.016	0.240

Notes: This table reports the top ten and bottom ten occupations ranked by the LLM-generated telework score. The third column gives the LLM-based telework probability, while the fourth column reports the corresponding NLP-generated telework probability for the same occupation. The NLP-generated probabilities are obtained from the latest occupation-level prediction file and are rounded to three decimal places.

C.3 Full regression tables for key results

Table 4: OLS Estimates of Telework Exposure and Paid Labor by Gender: Full Regression Results

	Women (1)	**Men (2)**
Dependent variable:	Paid labor	Paid labor
Telework exposure	0.0090***	0.0013
	(0.0035)	(0.0020)
District female LFPR	1.0511***	0.1013***
	(0.0364)	(0.0187)
Age	−0.0036***	−0.0112***
	(0.0002)	(0.0002)
Education (reference category: Not literate)		
Literate without formal schooling: EGS/NFEC/AEC	−0.0476	0.0137
	(0.0304)	(0.0268)
Literate without formal schooling: TLC	−0.0761*	−0.1225*
	(0.0388)	(0.0726)
Literate without formal schooling: Other	−0.0670***	0.0606***
	(0.0206)	(0.0205)
Below primary	−0.0063	0.0384***
	(0.0058)	(0.0048)
Primary	−0.0417***	0.0466***
	(0.0052)	(0.0037)
Middle	−0.0962***	0.0527***
	(0.0052)	(0.0036)
Secondary	−0.1182***	0.0205***
	(0.0058)	(0.0041)
Higher secondary	−0.2000***	−0.1642***
	(0.0079)	(0.0053)
Diploma/certificate course	0.0488***	0.0290***
	(0.0121)	(0.0078)
Graduate	−0.0518***	0.0016
	(0.0077)	(0.0041)
Postgraduate and above	0.0948***	0.0414***
	(0.0105)	(0.0054)

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Table 4: OLS Estimates of Telework Exposure and Paid Labor by Gender: Full Regression Results *(continued)*

	Women (1)	**Men (2)**
Dependent variable:	Paid labor	Paid labor
Marital status		
Never married	−0.0264	−0.1156
	(0.1623)	(0.1867)
Currently married	−0.0311	0.2526
	(0.1629)	(0.1866)
Widowed	−0.0505	0.0147
	(0.1628)	(0.1865)
Divorced/separated	0.1882	0.1721
	(0.1621)	(0.1869)
Religion (reference category: Hinduism)		
Islam	−0.0987***	−0.0088***
	(0.0070)	(0.0030)
Christianity	0.0135*	−0.0244***
	(0.0076)	(0.0054)
Sikhism	−0.0068	−0.0055
	(0.0071)	(0.0087)
Jainism	−0.0485***	0.0437***
	(0.0149)	(0.0102)
Buddhism	−0.0317**	−0.0175
	(0.0141)	(0.0112)
Zoroastrianism	0.0128	0.0086
	(0.0428)	(0.0426)
Other religions	0.0240	0.0169
	(0.0210)	(0.0124)
Social group (reference category: Scheduled Tribe)		
Scheduled Caste	−0.0677***	−0.0042
	(0.0086)	(0.0044)
Other Backward Class	−0.0563***	0.0082**
	(0.0083)	(0.0040)
Others	−0.0860***	0.0020
	(0.0089)	(0.0045)
Lagged log nighttime lights	−0.0173	−0.0049
	(0.0227)	(0.0076)
Constant	0.5520*	1.1455***
	(0.2839)	(0.2021)
District fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
State × year fixed effects	Yes	Yes
Survey weights	Yes	Yes
Observations	579,340	569,554
Clusters	600	600
R-squared	0.1322	0.2373
Within R-squared	0.0521	0.2265

Notes: The table reports the full set of coefficients from the OLS specifications summarized in main Table (4). Columns (1) and (2) report estimates for women and men, respectively. Both specifications use paid labor as the dependent variable and include district female labor-force participation, age, education, marital status, religion, social group, lagged nighttime lights, district fixed effects, year fixed effects, and state-by-year fixed effects. PLFS sampling weights are used in both regressions. Standard errors, clustered at the district level, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: IV Estimates of Telework Exposure and Women's Paid Labor: Full Regression Results

	(1) First stage	**(2) 2SLS**
Dependent variable:	Telework exposure	Paid labor
$TWO_d \times A_d^{ITM} \times 4G_t$	0.7146^{***}	
	(0.0682)	
Telework exposure		0.0396^{***}
		(0.0095)
District female LFPR	−0.2069	1.0428^{***}
	(0.2150)	(0.0375)
Age	0.0000	-0.0036^{***}
	(0.0001)	(0.0002)
Education (reference category: Not literate)		
Literate without formal schooling: EGS/NFEC/AEC	0.0464	−0.0493
	(0.0457)	(0.0300)
Literate without formal schooling: TLC	−0.0143	-0.0752^{*}
	(0.0430)	(0.0387)
Literate without formal schooling: Other	0.0315	-0.0682^{***}
	(0.0258)	(0.0206)
Below primary	−0.0017	−0.0064
	(0.0039)	(0.0058)
Primary	−0.0005	-0.0417^{***}
	(0.0030)	(0.0052)
Middle	0.0039	-0.0964^{***}
	(0.0032)	(0.0052)
Secondary	0.0042	-0.1184^{***}
	(0.0035)	(0.0058)
Higher secondary	0.0017	-0.2001^{***}
	(0.0034)	(0.0079)
Diploma/certificate	0.0078	0.0486^{***}
	(0.0088)	(0.0121)
Graduate	0.0017	-0.0518^{***}
	(0.0048)	(0.0077)
Postgraduate and above	0.0054	0.0947^{***}
	(0.0063)	(0.0105)

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Table 5: IV Estimates of Telework Exposure and Women's Paid Labor: Full Regression Results *(continued)*

	(1) First stage	**(2) 2SLS**
Dependent variable:	Telework exposure	Paid labor
Marital status		
Never married	0.1391	−0.0293
	(0.1497)	(0.1628)
Currently married	0.1441	−0.0342
	(0.1501)	(0.1634)
Widowed	0.1447	−0.0536
	(0.1502)	(0.1633)
Divorced/separated	0.1422	0.1851
	(0.1517)	(0.1626)
Religion (reference category: Hinduism)		
Islam	0.0023	−0.0988***
	(0.0053)	(0.0070)
Christianity	0.0111	0.0131*
	(0.0081)	(0.0076)
Sikhism	−0.0106	−0.0062
	(0.0136)	(0.0071)
Jainism	−0.0143	−0.0479***
	(0.0314)	(0.0152)
Buddhism	0.0246	−0.0320**
	(0.0169)	(0.0142)
Zoroastrianism	−0.1062**	0.0156
	(0.0450)	(0.0430)
Other religion	0.0051	0.0241
	(0.0282)	(0.0205)
Social group (reference category: Scheduled Tribe)		
Scheduled Caste	−0.0078	−0.0675***
	(0.0099)	(0.0086)
Other Backward Class	−0.0034	−0.0563***
	(0.0090)	(0.0084)
Other	−0.0076	−0.0858***
	(0.0096)	(0.0089)
Lagged log nighttime lights	0.2611**	−0.0357
	(0.1031)	(0.0246)
FSS signal × 4G growth	−0.7517	−0.0087
	(0.7110)	(0.0521)
District fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
State × year fixed effects	Yes	Yes
Survey weights	Yes	Yes
Kleibergen–Paap rk Wald F	109.85	
Kleibergen–Paap rk LM statistic	33.54	
Kleibergen–Paap LM p-value	< 0.001	
Anderson–Rubin Wald F		13.31
Anderson–Rubin p-value		0.0003
Observations	579,340	579,340

Notes: Column (1) reports the first-stage regression, with telework exposure as the dependent variable. Column (2) reports the corresponding two-stage least-squares regression for women's paid labor. Both specifications also include age, education, marital status, religion, social group, district fixed effects, year fixed effects, and state-by-year fixed effects. Standard errors, shown in parentheses, are clustered by district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: IV Estimates of Telework Exposure and Men's Paid Labor: Full Regression Results

	(1) First stage	**(2) 2SLS**
Dependent variable:	Telework exposure	Paid labor
$TWO_d \times A_d^{ITM} \times 4G_t$	0.7048^{***}	
	(0.0682)	
Telework exposure		0.0051
		(0.0052)
District female LFPR	−0.2190	0.1002^{***}
	(0.2183)	(0.0188)
Age	−0.0001	-0.0112^{***}
	(0.0001)	(0.0002)
Education (reference category: Not literate)		
Literate without formal schooling: EGS/NFEC/AEC	−0.0516	0.0137
	(0.0375)	(0.0267)
Literate without formal schooling: TLC	−0.0065	-0.1231^{*}
	(0.0402)	(0.0728)
Literate without formal schooling: Other	0.0210	0.0607^{***}
	(0.0226)	(0.0205)
Below primary	0.0051	0.0384^{***}
	(0.0041)	(0.0048)
Primary	0.0024	0.0466^{***}
	(0.0034)	(0.0037)
Middle	0.0009	0.0527^{***}
	(0.0035)	(0.0036)
Secondary	−0.0016	0.0205^{***}
	(0.0030)	(0.0041)
Higher secondary	0.0044	-0.1642^{***}
	(0.0037)	(0.0053)
Diploma/certificate	−0.0025	0.0290^{***}
	(0.0068)	(0.0078)
Graduate	0.0009	0.0016
	(0.0045)	(0.0041)
Postgraduate and above	0.0072	0.0414^{***}
	(0.0070)	(0.0055)

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Table 6: IV Estimates of Telework Exposure and Men's Paid Labor: Full Regression Results *(continued)*

	(1) First stage	**(2) 2SLS**
Dependent variable:	Telework exposure	Paid labor
Marital status		
Never married	0.2916***	−0.1166
	(0.0959)	(0.1869)
Currently married	0.2921***	0.2516
	(0.0959)	(0.1868)
Widowed	0.2860***	0.0138
	(0.0957)	(0.1867)
Divorced/separated	0.2707***	0.1713
	(0.0968)	(0.1870)
Religion (reference category: Hinduism)		
Islam	−0.0006	−0.0088***
	(0.0054)	(0.0030)
Christianity	0.0127	−0.0244***
	(0.0078)	(0.0054)
Sikhism	−0.0023	−0.0055
	(0.0135)	(0.0087)
Jainism	−0.0058	0.0438***
	(0.0324)	(0.0101)
Buddhism	0.0300*	−0.0175
	(0.0171)	(0.0112)
Zoroastrianism	−0.0658	0.0089
	(0.0482)	(0.0426)
Other religion	0.0099	0.0170
	(0.0308)	(0.0123)
Social group (reference category: Scheduled Tribe)		
Scheduled Caste	−0.0110	−0.0042
	(0.0103)	(0.0044)
Other Backward Class	−0.0023	0.0081**
	(0.0097)	(0.0041)
Other	−0.0050	0.0020
	(0.0104)	(0.0045)
Lagged log nighttime lights	0.2608**	−0.0076
	(0.1027)	(0.0081)
FSS signal × 4G growth	−0.7173	0.0473
	(0.7127)	(0.0455)
District fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
State × year fixed effects	Yes	Yes
Survey weights	Yes	Yes
Kleibergen–Paap rk Wald F	106.85	
Kleibergen–Paap rk LM statistic	33.62	
Kleibergen–Paap LM p-value	< 0.001	
Anderson–Rubin Wald F		0.95
Anderson–Rubin p-value		0.330
Observations	569,554	569,554

Notes: Column (1) reports the first-stage regression, with telework exposure as the dependent variable. Column (2) reports the corresponding two-stage least-squares regression for men's paid labor. Both specifications also include age, education, marital status, religion, social group, district fixed effects, year fixed effects, and state-by-year fixed effects. Standard errors, shown in parentheses, are clustered by district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.4 Full regression tables for heterogeneous effects

Table 7: Heterogeneous Effects by Marriage and Family Responsibilities: Full 2SLS Results

	Married (1)	**Young child (2)**	**Nuclear household (3)**
Telework exposure	−0.0261	0.0382***	0.0506***
	(0.0336)	(0.0118)	(0.0128)
Telework exposure × Currently married	0.0841**	–	–
	(0.0367)		
Telework exposure × Child aged 0–9	–	0.0231**	–
		(0.0104)	
Telework exposure × Nuclear household	–	–	0.0189**
			(0.0093)
Age	−0.0013***	−0.0017***	−0.0011***
	(0.0002)	(0.0002)	(0.0002)
District female LFPR	1.0883***	1.1255***	1.1320***
	(0.0416)	(0.0452)	(0.0487)
Education (reference category: Not literate)			
Literate without formal schooling: EGS/NFEC/AEC	−0.0525	−0.0563	−0.0729*
	(0.0341)	(0.0358)	(0.0406)
Literate without formal schooling: TLC	−0.0889**	−0.0829**	−0.0584
	(0.0408)	(0.0381)	(0.0405)
Literate without formal schooling: Other	−0.1031***	−0.0979***	−0.0915***
	(0.0231)	(0.0246)	(0.0262)
Below primary	−0.0197***	−0.0209***	−0.0230***
	(0.0063)	(0.0063)	(0.0070)
Primary	−0.0525***	−0.0543***	−0.0567***
	(0.0057)	(0.0057)	(0.0060)
Middle	−0.1024***	−0.1047***	−0.1048***
	(0.0057)	(0.0058)	(0.0061)
Secondary	−0.1238***	−0.1251***	−0.1265***
	(0.0066)	(0.0064)	(0.0066)
Higher secondary	−0.1995***	−0.1591***	−0.1562***
	(0.0087)	(0.0078)	(0.0081)
Diploma/certificate course	0.0486***	0.0072	−0.0048
	(0.0127)	(0.0143)	(0.0153)
Graduate	−0.0503***	−0.0935***	−0.0976***
	(0.0077)	(0.0084)	(0.0087)
Postgraduate and above	0.0980***	0.0419***	0.0339***
	(0.0107)	(0.0118)	(0.0125)

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Table 7: Heterogeneous Effects by Marriage and Family Responsibilities: Full 2SLS Results *(continued)*

	Married (1)	**Young child (2)**	**Nuclear household (3)**
Religion (reference category: Hinduism)			
Islam	−0.0975***	−0.1078***	−0.1080***
	(0.0073)	(0.0081)	(0.0084)
Christianity	0.0119	0.0118	0.0066
	(0.0086)	(0.0082)	(0.0087)
Sikhism	−0.0148*	−0.0151*	−0.0132
	(0.0080)	(0.0083)	(0.0090)
Jainism	−0.0471**	−0.0572***	−0.0680***
	(0.0190)	(0.0184)	(0.0236)
Buddhism	−0.0448***	−0.0434**	−0.0383**
	(0.0155)	(0.0170)	(0.0167)
Zoroastrianism	0.0303	0.0491	−0.0218
	(0.0440)	(0.0680)	(0.0831)
Other religions	0.0273	0.0219	0.0319
	(0.0243)	(0.0224)	(0.0246)
Social group (reference category: Scheduled Tribe)			
Scheduled Caste	−0.0724***	−0.0782***	−0.0822***
	(0.0093)	(0.0098)	(0.0104)
Other Backward Class	−0.0582***	−0.0615***	−0.0633***
	(0.0090)	(0.0095)	(0.0100)
Others	−0.0933***	−0.0984***	−0.1006***
	(0.0096)	(0.0102)	(0.0107)
Lagged log nighttime lights	−0.0477*	−0.0535**	−0.0535**
	(0.0261)	(0.0241)	(0.0263)
FSS signal × 4G growth	−0.0180	−0.0124	−0.0199
	(0.0576)	(0.0662)	(0.0677)
Currently married indicator	−0.0433***		
	(0.0122)	–	–
Child aged 0–9 indicator	–	−0.0234***	
		(0.0056)	–
Nuclear-household indicator	–	–	0.0237***
			(0.0041)
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes
Kleibergen–Paap rk Wald F	9.289	18.769	16.199
Kleibergen–Paap rk LM statistic	27.544	47.699	41.526
Kleibergen–Paap LM p-value	< 0.001	< 0.001	< 0.001
Observations	494,623	427,007	341,063
Clusters	600	600	600
Centered R^2	0.0412	0.0404	0.0410

Notes: The table reports the full second-stage coefficients corresponding to Panel A of Table (5). Column (1) compares currently married and never-married women. Column (2) is restricted to married women and compares women with and without a child aged 0–9. Column (3) is restricted to married women with a child aged 0–9 and compares nuclear and patrilocal households. All specifications include district, year, and state-by-year fixed effects, PLFS sampling weights, lagged nighttime lights, and the FSS signal interacted with 4G growth. Standard errors, shown in parentheses, are clustered at the district level. "–" denotes a regressor that is not included in that specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Heterogeneous Effects by Household Socioeconomic Conditions: Full 2SLS Results

	High MPCE (1)	**Social group (2)**	**Female-headed HH (3)**
Telework exposure	0.0220*	0.0496***	0.0261***
	(0.0130)	(0.0092)	(0.0080)
Telework exposure × High MPCE	0.0203**	–	–
	(0.0079)		
Telework exposure × Disadvantaged social group	–	−0.0611***	–
		(0.0150)	
Telework exposure × Female-headed household	–	–	0.0211*
			(0.0117)
Age	−0.0036***	−0.0037***	–
	(0.0002)	(0.0002)	
District female LFPR	1.0407***	1.0469***	1.0480***
	(0.0371)	(0.0373)	(0.0368)
Education (reference category: Not literate)			
Literate without formal schooling: EGS/NFEC/AEC	−0.0501*	−0.0543*	−0.0272
	(0.0299)	(0.0297)	(0.0285)
Literate without formal schooling: TLC	−0.0750*	−0.0630*	−0.0468
	(0.0385)	(0.0365)	(0.0379)
Literate without formal schooling: Other	−0.0690***	−0.0723***	−0.0404**
	(0.0206)	(0.0207)	(0.0198)
Below primary	−0.0071	−0.0097*	0.0231***
	(0.0058)	(0.0058)	(0.0060)
Primary	−0.0427***	−0.0459***	0.0044
	(0.0052)	(0.0053)	(0.0052)
Middle	−0.0977***	−0.1017***	−0.0319***
	(0.0052)	(0.0053)	(0.0046)
Secondary	−0.1202***	−0.1246***	−0.0507***
	(0.0059)	(0.0060)	(0.0050)
Higher secondary	−0.2025***	−0.2072***	−0.1052***
	(0.0080)	(0.0079)	(0.0053)
Diploma/certificate course	0.0454***	0.0409***	0.1414***
	(0.0121)	(0.0119)	(0.0120)
Graduate	−0.0552***	−0.0613***	0.0377***
	(0.0077)	(0.0078)	(0.0082)
Postgraduate and above	0.0910***	0.0844***	0.1780***
	(0.0106)	(0.0107)	(0.0105)

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Table 8: Heterogeneous Effects by Household Socioeconomic Conditions: Full 2SLS Results *(continued)*

	High MPCE (1)	**Social group (2)**	**Female-headed HH (3)**
Religion (reference category: Hinduism)			
Islam	−0.0986***	−0.0978***	−0.0788***
	(0.0070)	(0.0069)	(0.0069)
Christianity	0.0130*	0.0126	0.0085
	(0.0076)	(0.0079)	(0.0079)
Sikhism	−0.0072	−0.0096	−0.0163**
	(0.0071)	(0.0070)	(0.0072)
Jainism	−0.0485***	−0.0571***	−0.0783***
	(0.0152)	(0.0158)	(0.0149)
Buddhism	−0.0319**	−0.0509***	−0.0433***
	(0.0143)	(0.0144)	(0.0136)
Zoroastrianism	0.0160	0.0195	0.0159
	(0.0436)	(0.0432)	(0.0399)
Other religions	0.0248	0.0361	0.0286
	(0.0206)	(0.0228)	(0.0205)
Marital status			
Never married	−0.0335	−0.0381	
	(0.1616)	(0.1640)	–
Currently married	−0.0385	−0.0425	
	(0.1623)	(0.1646)	–
Widowed	−0.0576	−0.0614	
	(0.1621)	(0.1645)	–
Divorced/separated	0.1810	0.1764	
	(0.1614)	(0.1638)	–
Social group (reference category: Scheduled Tribe)			
Scheduled Caste	−0.0680***		−0.0747***
	(0.0086)	–	(0.0088)
Other Backward Class	−0.0572***		−0.0746***
	(0.0084)	–	(0.0086)
Others	−0.0872***		−0.1203***
	(0.0090)	–	(0.0089)
Lagged log nighttime lights	−0.0326	−0.0351	
	(0.0251)	(0.0271)	–
FSS signal × 4G growth	−0.0184	−0.0260	
	(0.0530)	(0.0544)	–
High-MPCE indicator	0.0076**		
	(0.0031)	–	–
Disadvantaged-social-group indicator	–	0.0152***	
		(0.0048)	–
Female-headed-household indicator	–	–	–
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes
Kleibergen–Paap rk Wald F	44.796	21.362	9.043
Kleibergen–Paap rk LM statistic	55.095	59.260	25.935
Kleibergen–Paap LM p-value	< 0.001	< 0.001	< 0.001
Observations	579,340	579,340	579,616
Clusters	600	600	600
Centered R^2	0.0512	0.0469	0.0364

Notes: The table reports the full second-stage coefficients corresponding to Panel B of Table (5). Columns (1) and (2) include age, district female LFPR, education, religion, marital status, lagged nighttime lights, and the FSS signal interacted with 4G growth. Standard errors, shown in parentheses, are clustered at the district level. "–" denotes a regressor that is not included in that specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Human Capital over the Life Cycle: Full 2SLS Results

	24–29 **(1)**	**30–34** **(2)**	**35–39** **(3)**	**40–44** **(4)**
Telework exposure	0.0410**	0.0589***	0.0450**	0.0146
	(0.0196)	(0.0221)	(0.0221)	(0.0252)
Telework exposure × Postsecondary education	−0.0856***	−0.0358	0.0597*	0.0049
	(0.0295)	(0.0288)	(0.0334)	(0.0274)
District female LFPR	0.8976***	1.1216***	1.1795***	1.0308***
	(0.0542)	(0.0803)	(0.0909)	(0.0799)
Age	0.0200***	0.0143***	0.0052**	−0.0066***
	(0.0016)	(0.0020)	(0.0021)	(0.0024)
Postsecondary-education indicator	0.0836***	0.0663***	0.0341***	0.0846***
	(0.0110)	(0.0112)	(0.0120)	(0.0127)
Religion (reference category: Hinduism)				
Islam	−0.0846***	−0.1201***	−0.1427***	−0.1264***
	(0.0109)	(0.0115)	(0.0159)	(0.0159)
Christianity	0.0164	0.0194	0.0359**	0.0428**
	(0.0208)	(0.0256)	(0.0181)	(0.0189)
Sikhism	0.0193	−0.0167	−0.0405**	−0.0088
	(0.0163)	(0.0209)	(0.0199)	(0.0356)
Jainism	−0.0267	−0.0580	−0.0962	0.0133
	(0.0427)	(0.0661)	(0.0614)	(0.0466)
Buddhism	−0.0677**	−0.0631*	−0.0372	−0.1034**
	(0.0272)	(0.0342)	(0.0317)	(0.0429)
Zoroastrianism	0.0608	0.0543	−0.1459*	0.2483
	(0.1162)	(0.1524)	(0.0783)	(0.1636)
Other religions	−0.0375	0.0711	0.0161	−0.0518
	(0.0352)	(0.0616)	(0.0580)	(0.0343)
Social group (reference category: Scheduled Tribe)				
Scheduled Caste	−0.0675***	−0.0648***	−0.0716***	−0.0894***
	(0.0151)	(0.0203)	(0.0187)	(0.0217)
Other Backward Class	−0.0540***	−0.0592***	−0.0674***	−0.1046***
	(0.0148)	(0.0198)	(0.0184)	(0.0195)
Others	−0.0711***	−0.0922***	−0.1160***	−0.1528***
	(0.0154)	(0.0201)	(0.0187)	(0.0221)

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Table 9: Human Capital over the Life Cycle: Full 2SLS Results *(continued)*

	24–29 **(1)**	**30–34** **(2)**	**35–39** **(3)**	**40–44** **(4)**
Marital status				
	−0.0464			0.5966***
Never married	(0.3183)	–	–	(0.0606)
	−0.3517	−0.2964***	−0.1744***	0.3995***
Currently married	(0.3182)	(0.0204)	(0.0230)	(0.0458)
	−0.0400	0.0822***	0.1715***	0.6793***
Widowed	(0.3181)	(0.0286)	(0.0264)	(0.0478)
	−0.0152	0.0067	0.0992***	0.6263***
Divorced/separated	(0.3188)	(0.0360)	(0.0364)	(0.0662)
	−0.0243	−0.1046***	−0.0531**	−0.0258
Lagged log nighttime lights	(0.0205)	(0.0297)	(0.0265)	(0.0401)
	−0.1495	−0.0507	0.0153	0.2556**
FSS signal × 4G growth	(0.1007)	(0.1202)	(0.1217)	(0.1076)
District fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes	Yes
Kleibergen–Paap rk Wald F	6.997	6.228	6.729	5.438
Kleibergen–Paap rk LM statistic	22.618	19.426	19.269	15.094
Kleibergen–Paap LM p-value	< 0.001	< 0.001	< 0.001	< 0.001
Observations	74,439	52,016	50,513	38,198
Clusters	597	587	592	579
Centered R^2	0.0988	0.0576	0.0500	0.0537

Notes: The table reports the full second-stage coefficients corresponding to Panel C of Table (5). Postsecondary education equals one for education levels 11–13 (diploma/certificate, graduate, and postgraduate or above) and zero for education levels 6–10; levels 1–5 are excluded from these regressions. All columns include age, district female LFPR, religion, social group, marital status, lagged nighttime lights, the FSS signal interacted with 4G growth, district fixed effects, year fixed effects, state-by-year fixed effects, and PLFS sampling weights. Standard errors, shown in parentheses, are clustered at the district level. "–" denotes a category that is omitted or empty in a particular estimation sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.5 Robustness Checks

C.5.1 Network Resolutions

Table 10: Sensitivity to Common-Reference Network Resolution

	20% Coarser (1)	**Baseline** (2)	**20% Finer** (3)
Telework exposure	0.0393***	0.0396***	0.0379***
	(0.0095)	(0.0095)	(0.0097)
Individual controls	Yes	Yes	Yes
District female LFPR	Yes	Yes	Yes
Lagged nighttime lights	Yes	Yes	Yes
FSS signal × 4G growth	Yes	Yes	Yes
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes
Kleibergen–Paap rk Wald F	103.71	109.85	84.85
Kleibergen–Paap rk LM p-value	< 0.001	< 0.001	< 0.001
Observations	579,340	579,340	579,340

Notes: The dependent variable is an indicator equal to one if the surveyed woman engages in paid labor. Each column reports a separate 2SLS specification using an alternative common-reference network resolution for constructing the radio-propagation instrument. Column (1) uses the 20% coarser network resolution, Column (2) reports the baseline specification, and Column (3) uses the 20% finer network resolution. All specifications include controls for age, education, marital status, religion, social group, district female labor-force participation, lagged nighttime lights, and the FSS signal interacted with 4G growth, as well as district fixed effects, year fixed effects, and state-by-year fixed effects. PLFS sampling weights are used throughout. Standard errors, clustered at the district level, are reported in parentheses. The Kleibergen–Paap rk Wald F-statistic and rk LM p-value are reported for instrument strength and identification. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: Sensitivity to Common-Reference Network Construction

Panel A. Receiver spacing (transmitter spacing fixed at 50 km)	15 km	30 km	40 km
	(1)	(2)	(3)
Telework exposure	0.0865***	0.0881***	0.0793***
	(0.0144)	(0.0143)	(0.0134)
FSS signal × 4G growth	−0.0487	−0.0210	0.0407*
	(0.0503)	(0.0195)	(0.0240)

Panel B. Transmitter spacing (receiver spacing fixed at 20 km)	30 km	40 km	60 km	70 km
	(1)	(2)	(3)	(4)
Telework exposure	0.0841***	0.0793***	0.0875***	0.0883***
	(0.0147)	(0.0151)	(0.0149)	(0.0147)
FSS signal × 4G growth	0.0175	−0.0039	0.0137	0.0305
	(0.0225)	(0.0142)	(0.0224)	(0.0381)

Panel C. Grid origin (spacings fixed at 50 km and 20 km)	TX shifted (25, 25)	RX shifted (10, 10)	Both shifted (25, 25); (10, 10)
	(1)	(2)	(3)
Telework exposure	0.0885***	0.0872***	0.0881***
	(0.0149)	(0.0144)	(0.0151)
FSS signal × 4G growth	−0.0322	−0.0315	−0.0660
	(0.1135)	(0.0294)	(0.1048)
Individual controls	Yes	Yes	Yes
Lagged nighttime lights	Yes	Yes	Yes
District, year, and state × year FE	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes
Observations	579,340	579,340	579,340

Notes: Each column reports a separate 2SLS specification that differs only in how the hypothetical common-reference transmitter and receiver network is constructed. Panel A varies receiver spacing holding transmitter spacing at 50 km; Panel B varies transmitter spacing holding receiver spacing at 20 km; Panel C holds both spacings at their baseline values (50 km and 20 km) and shifts the origin of the transmitter lattice, the receiver lattice, or both, with shifts in kilometers reported in the column headers. Every column uses the same estimation sample (579,340 observations and includes individual controls, lagged nighttime lights, district, year, and state-by-year fixed effects, PLFS survey weights, and the free-space signal measure interacted with national 4G growth. The baseline construction—50 km transmitter spacing, 20 km receiver spacing, and no origin shift—is the specification reported in Table **??** and is not repeated here; it yields a coefficient of 0.0846 (0.0138). Standard errors, clustered at the district level, are reported in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

C.5.2 Alternative Networks

C.5.3 Excluding COVID-19 years

Table 12: IV Estimates by Excluding COVID-19 Pandemic years (2020-21)

	(1) First stage	(2) 2SLS
Dependent variable:	Telework exposure	Paid labor
$TWO_d \times A_d^{ITM} \times 4G_t$	0.6986***	
	(0.0673)	
Exposure_{dt}		0.0838***
		(0.0150)
FSS signal × 4G growth	−0.0539	−0.1013
	(0.8139)	(0.0826)
Individual controls	Yes	Yes
Lagged nighttime lights	Yes	Yes
FSS signal × 4G growth	Yes	Yes
District fixed effects	Yes	Yes
State × year fixed effects	Yes	Yes
Survey weights	Yes	Yes
Kleibergen–Paap rk Wald F	107.69	
Kleibergen–Paap rk LM p-value	< 0.001	
Anderson–Rubin Wald F		29.46
Anderson–Rubin p-value		< 0.001
Observations	443,002	443,002

Notes: Column (1) reports the first-stage estimate. Column (2) reports the corresponding two-stage least-squares estimate for women's paid labor. All specifications control for the corresponding interaction constructed using free-space signal strength, lagged nighttime lights, and individual characteristics including age, education, marital status, religion, and social group. District and state-by-year fixed effects are included. PLFS sampling weights are used in all specifications, and standard errors, reported in parentheses, are clustered at the district level. The model is exactly identified. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.5.4 Additional Robustness Checks

Table 13: Additional Robustness Checks

	Without Lagged	**District**	**Sub-samples**	
	Female Labor Supply	**Representativeness**	**Ages 18–44**	**Ages 19–59**
	(1)	(2)	(3a)	(3b)
Telework exposure	0.0846***	0.0409***	0.0347***	0.0441***
	(0.0138)	(0.0099)	(0.0101)	(0.0101)
Lagged district female labor participation	—	1.0465***	1.0130***	1.1130***
		(0.0402)	(0.0435)	(0.0413)
FSS signal × 4G growth	−0.0200	−0.0090	−0.0655	−0.0251
	(0.0800)	(0.0553)	(0.0622)	(0.0575)
District-year observations > 50	No	Yes	No	No
Lagged female labor participation	No	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes
Lagged nighttime lights	Yes	Yes	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes	Yes
Survey weights	Yes	Yes	Yes	Yes
KP rk Wald F	106.27	101.88	109.67	107.27
Anderson–Rubin p-value	< 0.001	—	—	—
Observations	579,340	570,746	368,299	485,821

Notes: The table reports 2SLS estimates of the effect of telework exposure on women's paid labor. Column (1) reports the baseline specification without controlling for lagged district-level female labor-force participation. Column (2) restricts the sample to district-year cells with more than 50 observations. Columns (3a) and (3b) restrict the sample to women ages 18–44 and 19–59, respectively. Columns (2)–(3b) additionally control for lagged district-level female labor-force participation. Telework exposure is instrumented by the interaction of predetermined district teleworkability, terrain-adjusted predicted signal strength, and 4G growth. All specifications control for the corresponding free-space signal interaction, lagged nighttime lights, and individual characteristics, and include district, year, and state-by-year fixed effects. PLFS sampling weights are used. Standard errors in parentheses are clustered by district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Construction of Telework Scores using Supervised Text Classifier

D.1 Evidence-tag dictionary of phrases

Table 14: Evidence tags used to flag tasks typically requiring digital cues and physical presence

Tag	Type	
digital_cues	remote_compatible_cues	email / e-mail; cor
wc_violent_weekly	physical_presence_flag	
wc_outdoors_daily	physical_presence_flag	outdoor;
wc_disease_weekly	physical_presence_flag	
wc_minor_injuries_weekly	physical_presence_flag	weld / w
wc_walkrun_majority	physical_presence_flag	
wc_ppe_majority	physical_presence_flag	
gwa_general_physical_important	physical_presence_flag	manual work / manual labor; physically demanding; li

Table 14: Evidence tags used to flag tasks typically requiring digital cues and physical presence (continued)

Tag	Type	
gwa_move_objects_important	physical_presence_flag	load / loading; unload / unloading; lift / lifting; carry /
gwa_control_machines_important	physical_presence_flag	machine / machinery; pla
gwa_operate_vehicles_important	physical_presence_flag	driver / driving; ve
gwa_public_direct_important	physical_presence_flag	counter; front desk; reception; cashier
gwa_repair_mech_important	physical_presence_flag	repair; maintain
gwa_repair_elec_important	physical_presence_flag	repair / maintain / servic
gwa_inspect_important	physical_presence_flag	inspect / inspec

Notes: Phrase families are shown in readable form. In the implementation, each family is encoded as a regular-expression (regex) pattern to capture common variants (e.g., plurals, hyphenation, spacing). These evidence tags are used for interpretability and error audits; they do not mechanically determine the final teleworkability label.

D.2 NLP Terminology

Precision tells us how many results were actually positive out of those classified as positive by the model. The corresponding equation is:

$$\text{precision} = \frac{\text{True Positives}}{\text{True Positives+False Positives}} \tag{21}$$

Recall tells us how many of the positive cases the trained classifier accurately predicted in the data. The corresponding equation is:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives+False Negatives}} \tag{22}$$

Specificity tells us how well the model avoids false alarms. It is measured as:

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives+False Negatives}} \tag{23}$$

Sensitivity tells us how well the model correctly identifies truly positive cases. It is measured as:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives+False Negatives}} \tag{24}$$

F1 Score is the harmonic mean of Precision and Recall. It is calculated as:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision+Recall}} \tag{25}$$

F1 $\in$ [0,1]. If F1=1, it is the ideal case of no false positives and no false negatives. In practice, F1 above 0.9 is considered efficient.

Youden's J statistic selects the cutoff that maximizes the joint classification accuracy of true positives and true negatives.It is calculated as:

$$\mathbf{J} = \text{Sensitivity+Specificity-1} \tag{26}$$

Any value τ that maximizes eq. (26) is the optimal threshold choice.

D.2.1 Robustness Check: LLM-Based Teleworkability Scores

This appendix describes the procedure used to generate an alternative teleworkability measure using a large language model (LLM). The objective is to assess whether the occupation-level teleworkability patterns identified using the baseline NLP classifier are robust to an independently generated scoring approach.

Sampling of Occupations I randomly draw a sample of 400 occupations from the full NCO–2015 occupation set used in the baseline analysis. For each occupation, I use the occupation title and detailed description from the NCO manual as input text. These occupations are the same textual inputs used by the baseline NLP pipeline described in Section **??**. The sampling procedure ensures that the robustness exercise covers a diverse set of occupations across sectors and skill levels while keeping the computational cost of LLM evaluation manageable.

LLM Scoring Procedure Teleworkability probabilities are generated using a locally hosted `llama3.1:8b` model accessed through the Ollama inference framework. For each occupation, the model is prompted with the occupation title and description and instructed to return a single teleworkability probability between zero and one.

The prompt explicitly requires the model to return a strict JSON object with a single key `telework_prob`. The model is instructed to interpret the probability as follows:

- 0: the occupation cannot realistically be performed remotely,
- 1: the occupation can be fully performed remotely.

To ensure consistency across responses, several constraints are imposed in the prompting procedure:

1. The model is instructed to output probabilities with three decimal places.

2. The sampling temperature is set to 0.15 to reduce response variability.

3. The output format is restricted to JSON to facilitate automated parsing.

Each occupation is queried sequentially through the Ollama API. The returned responses are parsed using a custom extraction function that retrieves the teleworkability probability from the JSON output and constrains the value to the interval $[0, 1]$.

Handling of LLM Outputs The raw LLM responses occasionally contain formatting variations or coarse probability values. To address this, I implement a parsing procedure that extracts the probability value from the returned JSON string and rounds it to three decimal places. If the response contains a probability rounded to a single decimal (e.g., 0.8), the query is automatically repeated with a stricter formatting instruction.

All extracted probabilities are stored together with the occupation identifier and raw LLM response to allow verification and reproducibility.

Comparison with the Baseline NLP Measure The LLM-generated teleworkability scores are compared with the baseline NLP-based probabilities estimated in the main analysis. Two types of comparisons are performed.

First, I compute Pearson and Spearman correlations between the LLM-based and NLP-based teleworkability probabilities across the sampled occupations. These statistics measure the degree to which the two scoring methods rank occupations similarly.

Second, I evaluate both measures on the subset of occupations that overlap with the manually labeled gold-standard dataset used to train the baseline classifier. On this overlap sample, I compute the area under the receiver operating characteristic curve (ROC–AUC) and the area under the precision–recall curve (PR–AUC) for each scoring method.

These metrics provide a common benchmark for assessing how well each scoring procedure recovers the manually labeled teleworkability classification.

Implementation Details All LLM queries are executed locally using the Ollama inference server (`http://localhost:11434`). The robustness script is implemented in Python and performs the following steps:

1. Randomly sample occupations from the prediction dataset.
2. Query the LLM with the occupation title and description.
3. Parse the returned probability from the JSON response.
4. Store both the extracted probability and raw model output.
5. Compute correlation and classification metrics relative to the baseline NLP score and the gold-standard labels.

The complete code used to generate the LLM scores and evaluation metrics is provided in the replication package.